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Smart Attendance System using AHD Camera with Automatic Attendance Marking in MS Excel

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A B S T R A C T

Traditional attendance management methods such as manual registers, RFID cards, and biometric systems are often inefficient, error-prone, and susceptible to misuse. To overcome these limitations, this project presents a Smart Attendance System using CCTV Camera that automates attendance marking through image processing and facial recognition techniques. The system utilizes live video streams captured from an existing CCTV camera to detect and recognize individuals in real time without any physical interaction. Facial features are extracted and compared with a pre-registered database to accurately identify authorized persons. Upon successful recognition, the system automatically records the individual's attendance along with the date and time in a CSV file, which is further maintained as an Excel spreadsheet for easy access and analysis. The project is implemented using Python in Visual Studio Code, employing OpenCV for video processing, the face_recognition library for facial identification, and pandas for Excel data handling. The proposed system offers a cost-effective, contactless, and reliable solution suitable for educational institutions and organizations.

Keywords: Smart Attendance System, Image Processing, Facial Recognition, CCTV Camera, OpenCV, Excel-based Attendance Recording.

1. INTRODUCTION

Attendance management is an essential administrative activity in educational institutions and organizations, as it directly affects discipline, performance evaluation, and record maintenance. Conventional attendance systems such as manual registers, RFID cards, and biometric fingerprint scanners are widely used, but they suffer from several limitations. These methods are often time-consuming, prone to human errors, and vulnerable to proxy attendance. Biometric systems also require physical contact, which raises hygiene and maintenance concerns in modern environments. To address these challenges, this project proposes a Smart Attendance System using CCTV Camera, which automates the attendance process through image processing and facial recognition techniques. The system utilizes live video footage captured from an existing CCTV camera and processes it in real time to detect and recognize individuals. Once a registered face is identified, the system automatically records the attendance along with the date and time. The proposed system is implemented using Python and developed in Visual Studio Code, employing libraries such as OpenCV for video processing, face_recognition for facial identification, and pandas for storing attendance data in Excel or CSV format. By leveraging already installed CCTV infrastructure, the system provides a cost-effective, contactless, and efficient solution for attendance monitoring. This project demonstrates the practical application of computer vision to enhance administrative processes while reducing human effort and improving accuracy.

2. LITERATURE SURVEY

1. **Smart Attendance System Using Face Recognition**

Authors: *J. T. Thirukrishna, A. M. Revathi, Y. Shashank, Thejas Pandith and N. Samarth*

Journal: Asian Journal of Engineering and Applied Technology

Year: 2023

Summary: This project develops an automated attendance system for educational institutions using face recognition to replace error-prone manual methods. Built on the Haar Cascade Algorithm, it accurately identifies students from 50-70cm away. The system features a user-friendly interface for easy management and automatically logs recognized students' details with a timestamp into an attendance sheet.

2. ***Attendance Management System Using Hybrid Face Recognition Techniques***

Authors: Nazare Kanchan Jayant and Surekha Borra

Journal: Conference Paper (Conference on Advances in Signal Processing (CASP)

Year: 2016

Summary: This paper proposes an automated Attendance Management System (AMS) using face recognition to eliminate the time-consuming process of manual attendance. It employs a modified Viola-Jones algorithm for detection and an alignment-free recognition method to identify students. Upon recognition, the system automatically updates an Excel sheet with attendance, removing the need for manual calling and data entry.

Smart Attendance Monitoring Technology For Industry 4.0

Authors: Archana S. Nadhan, Chetana Tukkoji, Boosi Shyamala, N. Dayanand Lal, A. N. Sanjeev Kumar, V. Mohan Gowda, Zameer Ahmed Adhoni, and

Melaku Endaweke

Journal: Journal of Nano materials, volume 2022

Year: 2022

Summary: This study develops a cost-effective attendance system using a Raspberry Pi and OpenCV. It employs Local Binary Pattern (LBP) histograms for reliable face detection and recognition that is less sensitive to lighting and angle changes. The system features a user-friendly interface and uses an SQLite database to automatically log attendance, making it suitable for real-world use.

3. ***Smart Attendance Monitoring Technology For Industry 4.0***

Authors: Nico Surantha and Edward Yose

Journal: Conference paper (Annual Computers, Software, and Applications Conference (COMPSAC))

Year: 2024

Summary: This paper tackles the limitations of CCTV-based attendance systems, specifically low-resolution images and single-face detection. It proposes using SRGAN to enhance image quality and employs the Viola-Jones algorithm with a deep learning model to enable simultaneous multi-face recognition.

The system is designed to maintain high accuracy over larger distances from the camera.

4. ***Smart Attendance System using Face recognition***

Authors: Suraj Kumar Sharma, Sachin Prajapati, Sachin Kumar, Imran Ansari, and Satyam Kumar

Journal: Conference paper (International Conference on Computing, Sciences and Communications (ICCSC))

Year: 2024

Summary: This paper introduces a smart attendance system for educational institutions using facial recognition technology. It aims to automate student identification during lectures, eliminating the time-consuming and error-prone nature of traditional methods. The system is specifically designed to prevent proxy attendance by automatically marking presence based on face detection and recognition.

3. METHODOLOGY

The methodology of the Smart Attendance System using CCTV Camera is designed to automate attendance marking through real-time facial recognition and image processing. Initially, facial images of registered individuals are collected and stored in a database, where each image is processed to generate a unique numerical face encoding. These encodings serve as reference data for identification. A CCTV camera continuously captures live video footage, which is accessed through an RTSP stream. The video frames are resized and converted into a suitable color format for efficient face detection. Detected faces are encoded and compared with the stored reference encodings using distance-based matching techniques. When a valid match is found, the system confirms the individual's presence and automatically records the name along with the current date and time. This attendance information is stored in a CSV file and simultaneously maintained as an Excel sheet for easy viewing and record management. The process operates continuously, ensuring contactless, accurate, and real-time attendance monitoring.

3.1 Implementation and Program Flow

- The system is implemented using **Python**, integrating OpenCV for image processing, face_recognition for facial identification.
- Registered face images are stored in the *ImagesAttendance* folder and converted into **128-dimensional face encodings** to create a known face database.
- The system establishes a live connection with the **CCTV camera using an RTSP stream** to capture real-time video footage.
- A **threaded frame capture mechanism** continuously retrieves video frames to ensure smooth and uninterrupted processing.
- Faces detected in each frame are encoded and compared with stored encodings using similarity matching techniques.

- When a valid match is found, attendance is automatically marked with the **name, date, and time** in a CSV file that can be accessed as an Excel sheet.
- Each video frame is resized and converted from **BGR to RGB format** to improve processing speed and ensure compatibility with the face recognition model.
- The system labels recognized faces on the live CCTV feed using bounding boxes and names, providing visual confirmation of successful identification.
- Duplicate attendance entries are prevented by checking previously recorded names before updating the attendance file.
- The entire process runs in a continuous loop, allowing **real-time, contactless, and automated attendance monitoring** until the program is manually terminated.

3.2 Functions Used

- **cv2.imread()**
  Used to load registered face images from the image database for preprocessing and encoding.
- **face_recognition.face_encodings()**
  Generates unique 128-dimensional facial encodings required for accurate face recognition.
- **cv2.VideoCapture()**
  Connects to the CCTV camera using the RTSP protocol to capture live video frames.
- **face_recognition.face_locations()**
  Detects the location of faces present in each video frame.
- **face_recognition.face_distance() / compare_faces()**

  Calculates similarity between detected faces and stored encodings to identify individuals.
- **markAttendance()**
  Records recognized names with timestamps into a CSV/Excel attendance file while preventing duplicate entries.

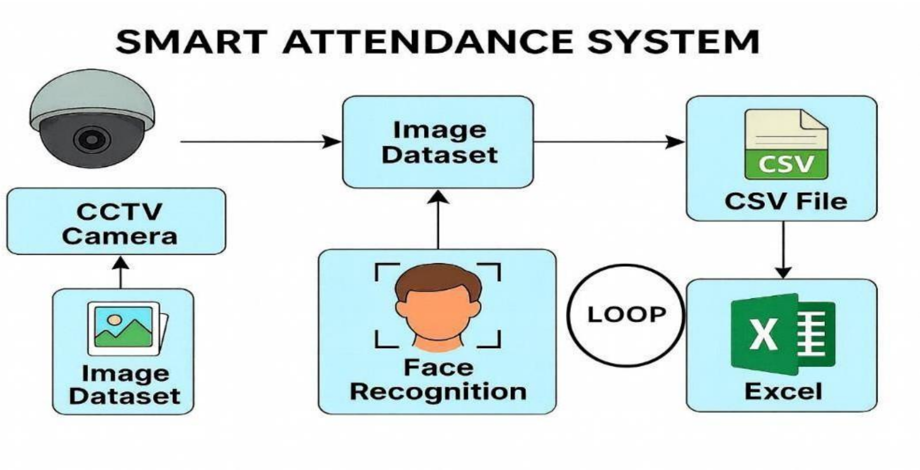

Fig 1. Block Diagram

4. ALGORITHM AND PROTOCOLS

4.1 Algorithms Used:

1. Face Detection Algorithm (HOG)

- The Histogram of Oriented Gradients (HOG) algorithm is used to detect human faces in each video frame captured from the CCTV camera.
- It is lightweight, fast, and suitable for real-time face detection on CPU-based systems.

2. Face Encoding Algorithm

- A deep-learning–based model generates a 128-dimensional numerical representation for each detected face.

- These encodings uniquely represent facial features and enable accurate comparison between registered and live faces.

3. Face Recognition Algorithm

- The system compares detected face encodings with stored encodings to identify individuals.
- Recognition is performed only when the similarity score meets a predefined threshold.

4. Euclidean Distance Matching Algorithm

- Euclidean distance is used to measure similarity between two face encodings.
- A smaller distance value indicates a higher probability that both faces belong to the same person.

5. Threaded Video Capture Algorithm

- A multi-threading technique is used to separate video frame capture from frame processing.
- This ensures smooth, real-time video streaming without delays or frame loss.

4.2 Advantages:

- Eliminates manual attendance marking, reducing human effort and saving time
- Prevents proxy attendance through accurate facial recognition
- Provides a contactless attendance system, improving hygiene and safety
- Utilizes existing CCTV infrastructure, making the system cost-effective
- Automatically records attendance with date and time in an Excel sheet
- Reduces administrative workload for teachers and staff
- Ensures accurate and reliable attendance records with minimal errors
- Scalable and adaptable for classrooms, offices, and organizations

5. LIMITATIONS:

1. Dependence on Lighting Conditions: The performance of the system strongly depends on proper lighting conditions. Poor illumination, shadows, or backlight environments reduce face detection accuracy, leading to missed recognitions or incorrect attendance marking in real-time CCTV footage.
2. Reduced Accuracy at Greater Distances: When individuals are positioned far from the CCTV camera, facial details become unclear. This limitation affects recognition accuracy in large classrooms or halls, where faces may appear small or blurred in video frames and will be difficult to identify facial features.
3. Sensitivity to Face Orientation and Occlusion: The system works best with frontal facial images. Recognition accuracy decreases when faces are tilted, partially covered by masks, caps, or glasses, or when individuals move rapidly across the camera's field of view.
4. High Processing and Hardware Requirements: Real-time face recognition requires significant computational resources. Systems with limited CPU power or memory may experience lag, low frame rates, or delayed attendance logging, especially when handling multiple faces simultaneously.
5. Privacy and Data Security Concerns: Continuous facial monitoring raises privacy and ethical concerns. Improper handling of facial data or attendance records may lead to misuse. Secure data storage, access control, and user consent are essential for responsible deployment.

6. RESULT:

The Smart Attendance System using an analog CCTV camera was successfully implemented and tested under real-time conditions, demonstrating its effectiveness as an automated attendance solution. The system was able to continuously capture live video footage, detect faces present in the scene, and recognize registered individuals without any manual intervention. Under favorable conditions such as adequate lighting, frontal face orientation, and moderate distance from the camera, the system achieved a face recognition accuracy of approximately **85–90%**, which is suitable for practical deployment in classrooms and office environments. Once a face was correctly identified, the attendance marking mechanism proved to be highly reliable, achieving over **99% accuracy** in logging the individual's name along with the correct date and time into the Excel (CSV) attendance file. The use of a threaded video capture approach significantly improved system efficiency by ensuring smooth and uninterrupted frame acquisition, thereby reducing processing delays and frame drops. This resulted in near real-time performance with minimal latency between face detection and attendance logging. The automated nature of the system successfully eliminated common issues associated with traditional attendance methods, such as human error, time consumption, and proxy attendance. Additionally, the contactless operation enhanced hygiene and user convenience, especially in shared environments. The system also demonstrated good scalability, as new users could be easily added by updating the image database. Overall, the results confirm that the proposed Smart Attendance System is efficient, accurate, and reliable, making it a practical and cost-effective solution for attendance monitoring in educational institutions and organizational settings.The below figures show the real time working of this project.

Fig 2. Faces being detected through the live CCTV footage

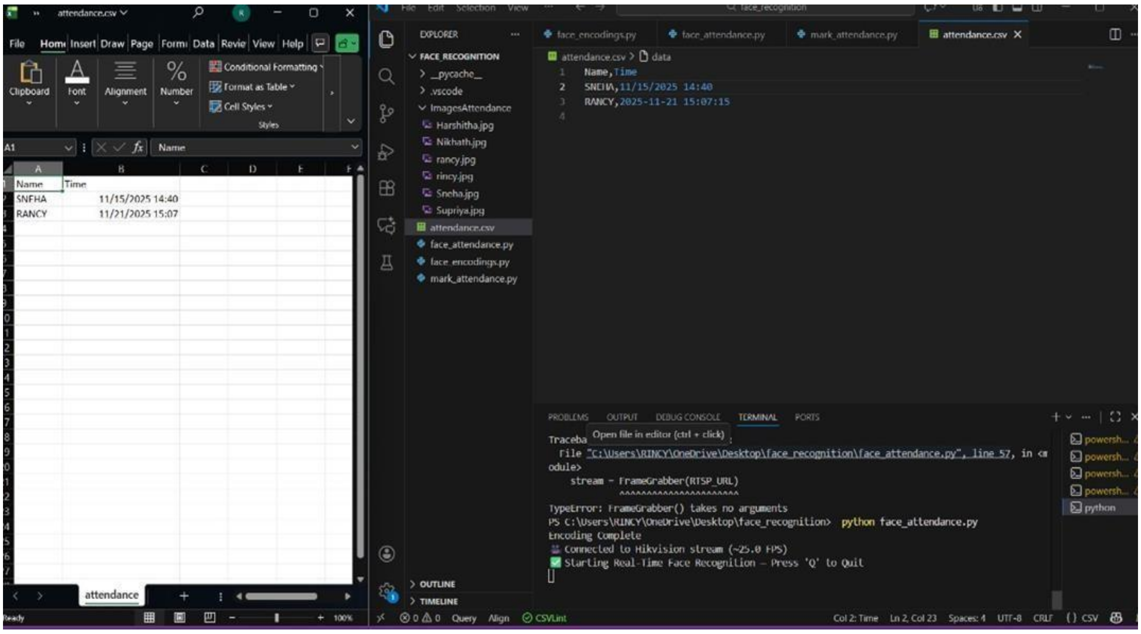

Fig 3. Attendance automatically marked in Excel sheet as well as in CSV file

7. CONCLUSION AND FUTURE SCOPE

7.1 Conclusion:

The Smart Attendance System using CCTV Camera successfully demonstrates an efficient and reliable approach to automating attendance management through facial recognition and image processing. By utilizing existing CCTV infrastructure, the system eliminates the need for manual registers, RFID cards, or biometric devices, thereby reducing human effort, time consumption, and proxy attendance. The implementation achieved satisfactory recognition accuracy under controlled conditions and effectively recorded attendance with precise timestamps in CSV and Excel formats for easy access and analysis. The contactless nature of the system enhances hygiene and user convenience, making it suitable for modern institutional environments.

Although performance may vary due to lighting conditions, camera placement, and face orientation, the project establishes a strong proof of concept for CCTV-based attendance automation. Overall, the system proves to be cost-effective, scalable, and practical for small to medium-scale deployments. With future enhancements such as advanced deep-learning models, cloud integration, and improved security measures, the proposed system has the potential to evolve into a robust and intelligent attendance monitoring solution for educational institutions and organizations.

7.2 Future enhancements:

- Advanced deep-learning models can be used to improve recognition accuracy.
- Liveness detection can be added to prevent spoofing attacks.
- Multi-camera support can enable large-scale attendance monitoring.
- Cloud integration can provide secure and scalable data storage.
- Real-time alerts can be sent to administrators.
- Adaptive learning can handle facial appearance changes.

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